

HRI - Observation Unit Robotic Kiosk

Problem Space Co-Design Canvas - Week 9 Interim

User group(s)

- Primary: Observation Unit nurses
- Secondary: patients, porters, biomedical/IT technicians
- Oversight: physicians, Infection Prevention, governance and facilities teams

Characteristics

- Mixed expertise and comfort with robots
- Patients may be anxious, unwell, multilingual or physically limited
- Staff work in a noisy shared environment with limited space

Needs

- A stationary, predictable and non-threatening system
- Reliable vitals capture with clear sensor-failure handling
- Nurse control over every classification and escalation
- Easy disengagement, privacy and physical clearance

User goals

Short term

- Capture selected vital signs consistently
- Route a clear summary to the nurse
- Reduce repeated manual transcription where safe

Long term

- Assess whether robot-assisted intake adds value without distress
- Build trust through transparency and safe degradation
- Determine whether future physical integration is justified

ROBOTIC SYSTEM task(s)

Short term

- Guide patient positioning for approved sensors
- Stream device readings and flag invalid/missing signals
- Display staff-facing recommendation only after nurse review
- Stop safely and request manual assessment when uncertain

Long term

- Remain stationary during pilot
- Log sensor failures, overrides and interruptions
- Avoid autonomous movement, diagnosis or treatment

Advantages / why this setting may benefit

Direct device data capture

Consistent prompts

Physical presence can guide positioning

Can operate beside a vitals station

Creates clear opportunities for fail-safe design